

Label-Efficient Emotion and Sentiment Analysis

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ABSTRACT

Emotion and sentiment analysis (ESA) assists machines to serve humans more intelligently. However, collecting large-scale high-quality datasets for training ESA models in a supervised manner is expensive, time-consuming, and difficult in practice. This tutorial focuses on the label-efficient ESA (LeESA) learning methods. Specifically, we first introduce the stimuli and characteristics of emotion and then illustrate seven typical training paradigms, followed by applications and future directions of LeESA.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → **Sentiment analysis**.

KEYWORDS

Affective Computing, Emotional Intelligence, Emotion and Sentiment Analysis, Label-Efficient Learning

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1 TUTORIAL MOTIVATION

ESA is crucial for machine intelligence [4]. In the past two decades, ESA has made notable progress on benchmark, affective feature extraction, and recognition methodology [10]. Hence, ESA has been applied to various real-world scenarios such as mental health, tourism services, and advertising recommendations. Nevertheless, due to the subjectivity, complexity, uncertainty and ambiguity, and subtlety of emotion and sentiment [7, 8], collecting large-scale, high-quality data for training ESA models remains expensive, time-consuming, and difficult in practice.

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First, **subjectivity** refers to the fact that for given affective stimuli, the emotional reactions across viewers might be different [9]. Second, **complexity** denotes that emotion is becoming diverse and fine-grained [2]. No consensus on how many emotion categories are necessary has been reached among psychologists [10]. More importantly, among the fine-grained emotions, there are many compound ones [3], which makes it difficult to collect sufficient data for all the categories. Third, **uncertainty and ambiguity** means that the emotions the same content expresses might be uncertain and ambiguous [6]. Taking a photo with “What wonderful weather” as an example, we may infer that it is a sunny day. However, if a photo about a storm is attached, it is more likely to use sarcasm in textual content to reflect the bad weather [5]. Finally, **subtlety** denotes the micro-characteristic of affective stimuli, especially micro-expressions [1].

Inspired by these characteristics of emotion and sentiment, this tutorial aims to discuss the label-efficient ESA (LeESA) learning methods. To sum up, we focus on three questions of ESA, i.e., How to perform ESA with noisy, limited, or even no labels? How to deal with incremental emotion categories and training samples? How to increase the transferability and generalizability of the trained models to new and unseen domains?

2 TUTORIAL CONTENT

The tutorial is planned for a half-day and organized into 10 sections. The topics and schedule are displayed in the following Table.

Section	Duration	Speaker
(1) Background and introduction	20 min	Sicheng Zhao
(2) Low-shot ESA, (3) Incremental ESA	30 min	Xiaopeng Hong
(4) Weakly-supervised ESA, (5) Semi-supervised ESA	30 min	Guoli Jia
(6) Domain adaptive ESA, (7) Domain generalizable ESA	30 min	Sicheng Zhao
(8) Unsupervised ESA	20 min	Yanyan Zhao
(9) Applications, (10) Future directions	30 min	Jianhua Tao

Next, we briefly summarize the content of each section. Particularly, given X_S , Y_S , X_T , Y_T denote the sets of samples from source domain, labels from source domain, samples from target domain, and labels from target domain respectively, we illustrate seven typical paradigms of LeESA.

Background and introduction. The tutorial starts with the introduction of stimuli signals and characteristics of traditional supervised ESA, sequentially illustrating the motivation and methodologies of LeESA.

Low-shot ESA. X_T and Y_T are available during training, and some categories only contain a few or no samples.

Incremental ESA. X_T and Y_T are available during training, and some categories and/or X_T are incremental.

Weakly-supervised ESA. X_T and unclear Y_T are available during training.

Semi-supervised ESA. X_T and a few Y_T are available during training, and $|Ava(Y_T)| \ll T$, where $Ava(\cdot)$ denotes the function to find available instances.

Domain adaptive ESA. X_S , Y_S , and X_T are available during training.

Domain generalizable ESA. X_S and Y_S are available during training.

Unsupervised ESA. X_T is available during training.

Applications. ESA supports various real-world applications. Furthermore, we discuss the typical applications of LeESA methods which alleviate the challenge of limited resources.

Future directions. Although remarkable progress has been made on LeESA, there are still several unsolved open issues. Therefore, we discuss the promising future directions from three aspects: more practical settings, new methodologies for LeESA, and ethical and legal restrictions.

3 TUTORIAL ORGANIZERS

Sicheng Zhao is a research associate professor at Tsinghua University. He worked as a postdoc research scientist at Columbia University from 2020 to 2022, as a postdoc research fellow at the University of California Berkeley from 2017 to 2020, and as a postdoc at Tsinghua University from 2016 to 2017. He obtained his Ph.D. degree at Harbin Institute of Technology in 2016. During his Ph.D., he visited the National University of Singapore as a research intern from 2013 to 2014. His research interests include affective computing, multimedia, and machine learning. He is a senior member of the ACM, IEEE, CCF, and CSIG. He is an associate editor of IEEE TIP and IEEE TAFRC, a leading guest editor of IEEE TFS and ACM TOMM, a program chair of the 1st CSIG Conference on Emotional Intelligence, an area chair or senior program committee of NeurIPS, ACM MM, ECCV, AAAI, and IJCAI, and a leading special session organizer of ICMR, ICASSP, ICME, and ICIP.

Guoli Jia is a PhD student at Tsinghua University, supervised by Prof. Bowen Zhou. He obtained his master's degree at Nankai University in 2024, advised by Prof. Jufeng Yang. His research interests include affective computing, controllable generation, and semi-supervised learning. Guoli Jia has published 10 papers in top-tier journals and conferences, such as IEEE SPM, IEEE TAFRC, ACM MM, CVPR, and ECCV. He was selected as an outstanding reviewer of CVPR 2024.

Xiaopeng Hong is a professor at Harbin Institute of Technology (HIT), P. R. China. He received his Ph.D. degree in computer application technology from Harbin Institute of Technology, P. R. China, in 2010. He had been a distinguished research fellow at Xi'an Jiaotong University, P. R. China, and an adjunct professor at the University of Oulu, Finland. His current research interests include incremental learning, visual surveillance, and micro-expression analysis. He has served as an area chair/SPC for ACM MM 2020-2023, AAAI 2020-2022, and IJCAI 2021, 2022, 2024, ICME 2021-2024, an associate

editor/guest editor for the journal of Image and Vision Computing, Pattern Recognition Letter, Signal, Image and Video Processing, and Frontiers, and a co-organizer for eight international workshops in conjunction with IEEE CVPR, ACM MM, IEEE FG. He is a senior member of IEEE.

Yanyan Zhao, PhD, a Professor and Doctoral Supervisor of Faculty of Computing at Harbin Institute of Technology and a Visiting Scholar of University of California, Berkeley. She received her Ph.D. degree in computer application technology from Harbin Institute of Technology, P. R. China, in 2011. Her research interests include affective computing and social computing. In recent years, she has presided over a number of foundation projects. She was awarded the first prize of Heilongjiang Provincial Scientific and Technological Progress, and the second prize of Heilongjiang Provincial Technological Invention.

Jianhua Tao received the M.S. degree from Nanjing University, Nanjing, China, in 1996, and the Ph.D. degree from Tsinghua University, Beijing, China, in 2001. He is currently a Professor with the Department of Automation, Tsinghua University. He is a recognized scholar in the fields of speech and language processing, multimodal human-computer interaction, and affective computing. He is a Fellow of the China Computer Federation (CCF), elected in 2017 "for contributions to spoken language processing and affective computing". He is a senior member of the IEEE. He was elected Chairperson of the ISCA Special Interest Group on Chinese Spoken Language Processing (ISCA SIG-CSLP) (2019-2020) and the Technical Program Chair of INTERSPEECH 2020. He was also an elected member of the Executive Committee of AAAC association (2007-2017) and served on the Steering Committee of IEEE Transactions on Affective Computing (2009-2017). He currently serves as the Subject Editor of Speech Communication and the Editorial Board Member of Journal on Multimodal User Interfaces and Journal of Computer Research and Development.

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